



UNIVERSITAT DE
BARCELONA

Facultat de Matemàtiques
i Informàtica

M.Sc. Thesis

Fundamental Principles of Data Science

Explaining Word Interactions Using Integrated Directional Gradients

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INTRODUCTION

Motivation

- **Deep Neural Networks (DNNs)**

Excellent ability to capture high-order interactions between features.

- **Linguistic Compositionality**

Meaning of a complex expression = meanings of its constituents + rules used to combine them.

INTRODUCTION

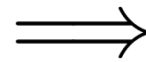
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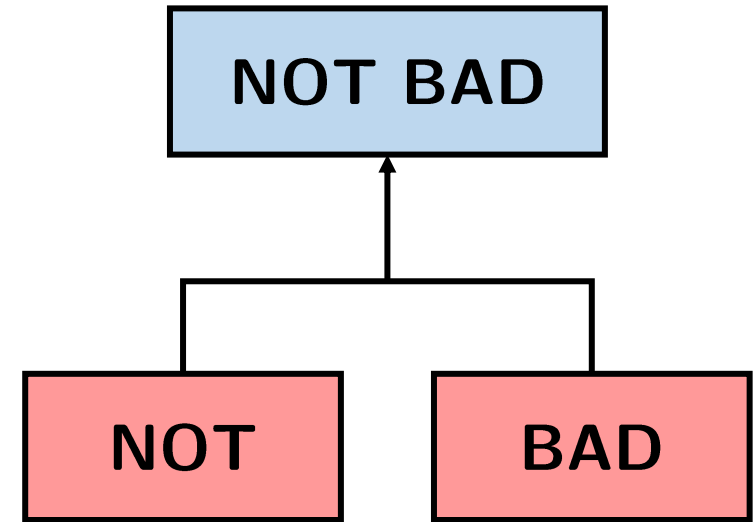


To better understand complex text models, we need explainability techniques that can **quantify the effect of feature interactions**.

INTRODUCTION

Motivation

To better understand complex text models, we need explainability techniques that can **quantify the effect of feature interactions**.



INTRODUCTION

Previous Work

- **Traditional feature importance explainability methods**

- Local Interpretable Model-agnostic Explanations (LIME)
- Shapley Additive Explanations (SHAP)
- Integrated Gradients (IG)

Only explain the effect
of single features
(1st order explanations)

INTRODUCTION

Previous Work

- **High-order Approaches** → Can explain feature interactions (high-order explanations)

- Contextual Decomposition (CD)
- Agglomerative CD (ACD)
- SHAP interaction values
- HEDGE
- Integrated Hessians (IH)

Potential Limitations:

- Not applicable to attention-based models.
- Computationally intensive.
- Limited to 2nd order interactions.

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Potential Limitations:

- Not applicable to attention-based models.
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- **Integrated Directional Gradients (IDG):**

- ✓ Designed for differentiable DNN functions.
- ✓ Computationally efficient.
- ✓ Can explain arbitrary order dependencies.

INTRODUCTION

Goals

1. **Theoretically explore IDG and discuss the particularities of its application to text models.**
2. **Explore and define metrics to assess the performance of IDG, and compare it to other methods.**
3. **Apply and evaluate IDG on a hate speech detection model.**

INTEGRATED DIRECTIONAL GRADIENTS

The Feature Group Attribution Problem

Axiomatic Requirements

The IDG Solution

Applying IDG to Text Models

INTEGRATED DIRECTIONAL GRADIENTS

IDG is aimed at solving the Feature Group Attribution Problem.

INTEGRATED DIRECTIONAL GRADIENTS

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Given: $\mathcal{X} \equiv$ feature space, $\mathcal{Y} \equiv$ target space

- **Input features:** $x = (x_1, \dots, x_n) \in \mathcal{X}$
- **Baseline:** $b = (b_1, \dots, b_n) \in \mathcal{X}$
 - ↳ Reference instance to compute importances.
Should represent absence of information.
- **Predictor:** $f : \mathcal{X} \rightarrow \mathcal{Y}$
- **Family of Meaningful Subsets:**
 $M \in \mathcal{P}(\{x_1, \dots, x_n\})$
 - ↳ Combinations of features that represent meaningful parts of the input.

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Assign to every meaningful subset of features an **importance score**.

$$v : M \longrightarrow \mathbb{R}$$
$$S \longmapsto v(S)$$

$v(S)$ should be a proxy of the importance of S to the predictor, with respect to the baseline b .

INTEGRATED DIRECTIONAL GRADIENTS

Axiomatic Requirements for a Solution of the Feature Group Attribution Problem

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- (1) **Non-negativity:** Feature importance should be a proxy of the amount of information (statistical entropy) that is used by the model, thus a non-negative quantity.

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- (4) **Superadditivity:** the value of the disjoint union of two feature subsets is greater than or equal to the sum of values of the two subsets.

INTEGRATED DIRECTIONAL GRADIENTS

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- (7) **Symmetry preservation:** features that are functionally and structurally equivalent, apportion the same amount of importance to all feature subsets there may be included in.

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- (7) **Symmetry preservation:** features that are functionally and structurally equivalent, apportion the same amount of importance to all feature subsets there may be included in.
- (8) **Implementation invariance:** as long as the predictor function is the same, feature importance does not depend on the underlying DNN architecture.

INTEGRATED DIRECTIONAL GRADIENTS

The IDG Solution

- Inspired by
 - **Integrated Gradients (IG)** → DNN Explainability.
 - **Harsanyi Dividends** → Game Theory.
 - ↳ Value uniquely attributable to the cooperation of features

INTEGRATED DIRECTIONAL GRADIENTS

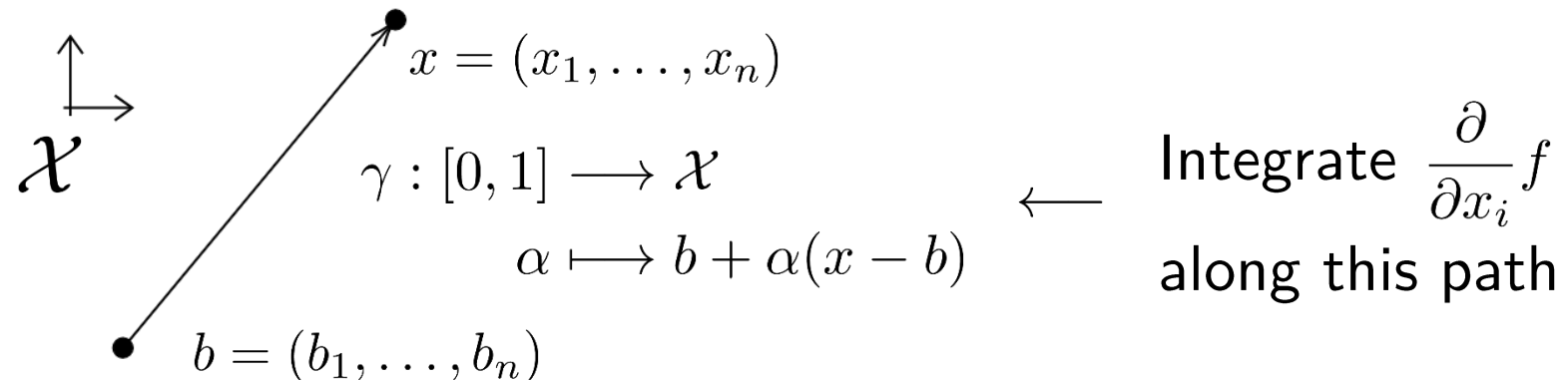
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 - $\hookrightarrow$ Value uniquely attributable to the cooperation of features

- Idea

- **Dividend of a Single Feature:** $\sim \left| \int_0^1 \frac{\partial}{\partial x_i} f(b_i + \alpha(x_i - b_i)) \, d\alpha \right|$



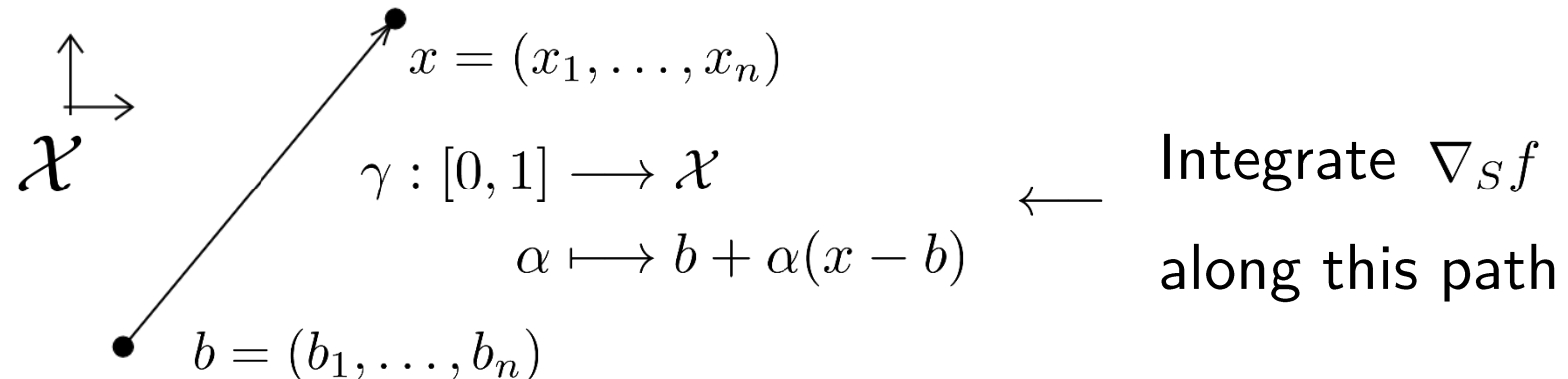
INTEGRATED DIRECTIONAL GRADIENTS

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- Idea

- **Dividend of a Single Feature:** $\sim \left| \int_0^1 \frac{\partial}{\partial x_i} f(b_i + \alpha(x_i - b_i)) \, d\alpha \right|$
- **Dividend of a Group of Features:** $\sim \left| \int_0^1 \nabla_S f(b_i + \alpha(x_i - b_i)) \, d\alpha \right|$

$\nabla_S f \equiv$ Directional derivative in the components given by S



INTEGRATED DIRECTIONAL GRADIENTS

The IDG Solution

• **Formalization:** given $S \in M$

- **Integrated Directional Gradient:** $\text{IDG}(S) = \int_0^1 \nabla_S f(b_i + \alpha(x_i - b_i)) \, d\alpha$
- **Dividend:** $d(S) = \frac{|\text{IDG}(S)|}{\sum_{T \in M} |\text{IDG}(T)|}$
- **Importance Score:** $v(S) = [\text{sign}(\text{IDG}(S))] \cdot \sum_{\substack{T \subseteq S \\ T \in M}} d(T)$

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Theorem

The importance function of IDG satisfies the axiomatic requirements (1) – (8).

INTEGRATED DIRECTIONAL GRADIENTS

Applying IDG to Text Models

- **Requirements:** define a well-suited **baseline** and **family of meaningful subsets**.

INTEGRATED DIRECTIONAL GRADIENTS

Applying IDG to Text Models

- **Requirements:** define a well-suited **baseline** and **family of meaningful subsets**.
 - **Baseline:** should represent the absence of any meaningful contributions.

Several approaches exist: [MASK], [PAD], [UNK], 0, mean of input embeddings...

Comprehensive studies suggest that a vector full of 0's is the most suitable option.

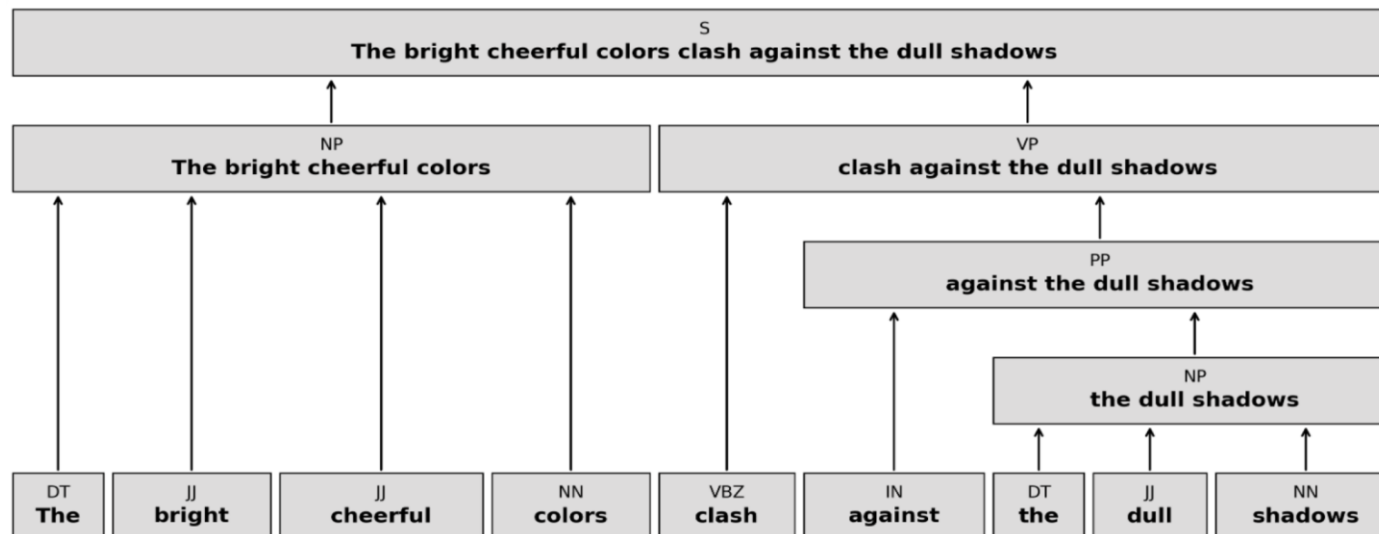
$$b = (0, \dots, 0) \begin{cases} \rightarrow b_{\text{BERT}} = ([\text{PAD}], \dots, [\text{PAD}]) \\ \rightarrow b_{\text{XLNet}} = ([\text{UNK}], \dots, [\text{UNK}]) \end{cases}$$

INTEGRATED DIRECTIONAL GRADIENTS

Applying IDG to Text Models

- **Requirements:** define a well-suited **baseline** and **family of meaningful subsets**.
 - **Family of Meaningful Subsets:** should restrict explanations to combinations that agree with the syntactic and semantic structure of language.

Constituency Parse Trees: hierarchical structures that represent the syntactic structure of a sentence.



EVALUATION METRICS

Word-level Attributions

High-order Explanations

Global Qualitative Evaluation

EVALUATION METRICS

Word-level Attributions: evaluate the quality of attributions for single words.

We will evaluate two dimensions of explanation quality:

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- **Plausibility:** agreement with human rationales.
 - ↳ How well the importance scores produced by the explainability method align with human-annotated scores.

EVALUATION METRICS

Word-level Attributions: evaluate the quality of attributions for single words.

We will evaluate two dimensions of explanation quality:

- **Plausibility:** agreement with human rationales.
 - ↳ How well the importance scores produced by the explainability method align with human-annotated scores.
- **Faithfulness:** agreement with internal model behaviour.
 - ↳ How well the importance scores provided by the explainability method reflect the functional dependencies of the model.

EVALUATION METRICS

Word-level Attributions

- **Plausibility:** agreement with human rationales.

We will use two metrics to evaluate plausibility:

- **agreement@k:** measures how well does the explainability method agree with humans in ranking the top k most important features.

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Word-level Attributions

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- **Area Under the Precision-Recall Curve (AUPRC):** measures how well the method identifies any token that humans consider important, regardless of its ranking.

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Word-level Attributions

- **Plausibility:** agreement with human rationales.

We will use two metrics to evaluate plausibility:

- **agreement@k:** measures how well does the explainability method agree with humans in ranking the top k most important features.
 - ↳ Focus on **ordinal agreement**.
- **Area Under the Precision-Recall Curve (AUPRC):** measures how well the method identifies any token that humans consider important, regardless of its ranking.
 - ↳ Focus on raw **identification ability**.

EVALUATION METRICS

Word-level Attributions

- **Faithfulness:** agreement with true model functional dependency.

We will use two metrics to evaluate faithfulness:

- **Comprehensiveness:** drop in model performance when important tokens are masked by a baseline value.

‘Are the tokens marked as important really important?’

↳ $\uparrow$ Comprehensiveness $\Rightarrow$ $\uparrow$ Faithfulness.

EVALUATION METRICS

Word-level Attributions

- **Faithfulness:** agreement with true model functional dependency.

We will use two metrics to evaluate faithfulness:

- **Sufficiency:** drop in model performance when non-important tokens are masked by a baseline value.

‘Do the tokens marked as contain enough information?’

↳ ↓ Sufficiency $\Rightarrow$ ↑ Faithfulness.

EVALUATION METRICS

Word level Attributions

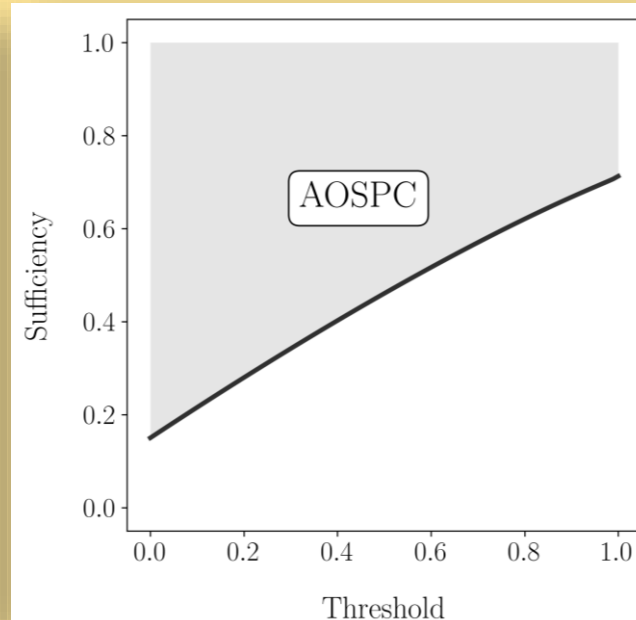
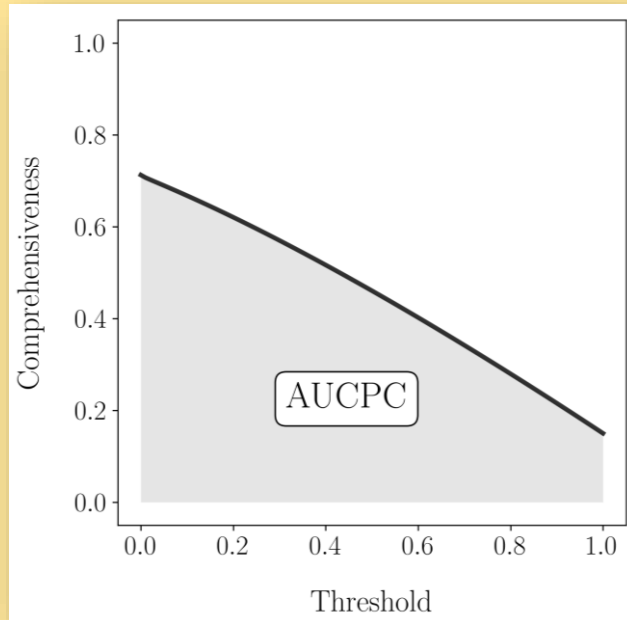
Comprehensiveness and sufficiency **rely on a fixed threshold** to define 'feature importance'.

EVALUATION METRICS

Word level Attributions

Comprehensiveness and sufficiency **rely on a fixed threshold** to define 'feature importance'.

Solution: construct {comprehensiveness, sufficiency}-threshold curves by evaluating the metrics across all possible thresholds.



AUCPC: Area Under the Comprehensiveness Perturbation Curve

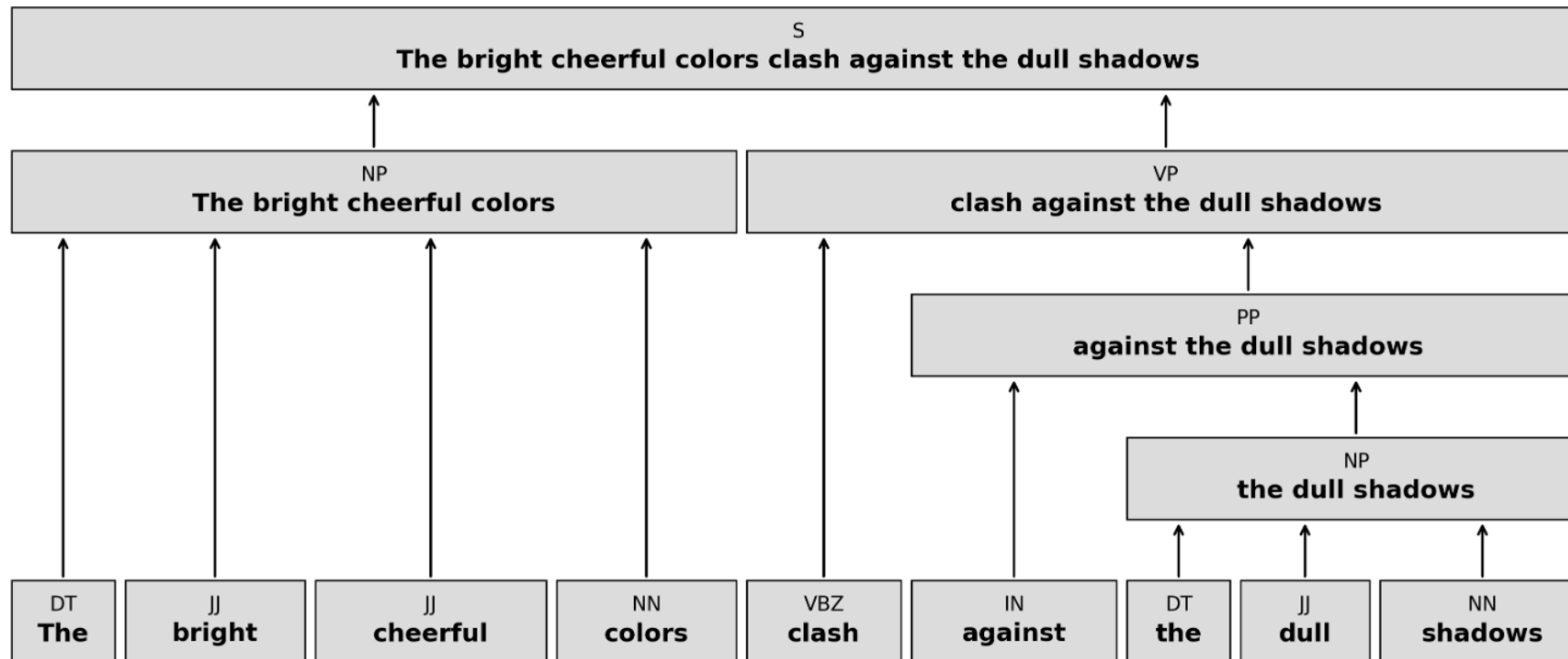
AOSPC: Area Over the Sufficiency Perturbation Curve

EVALUATION METRICS

High-order Attributions

We will evaluate high-order attributions from the **faithfulness** perspective.

Step 1: use a graph algorithm to depth-balance the constituency tree.

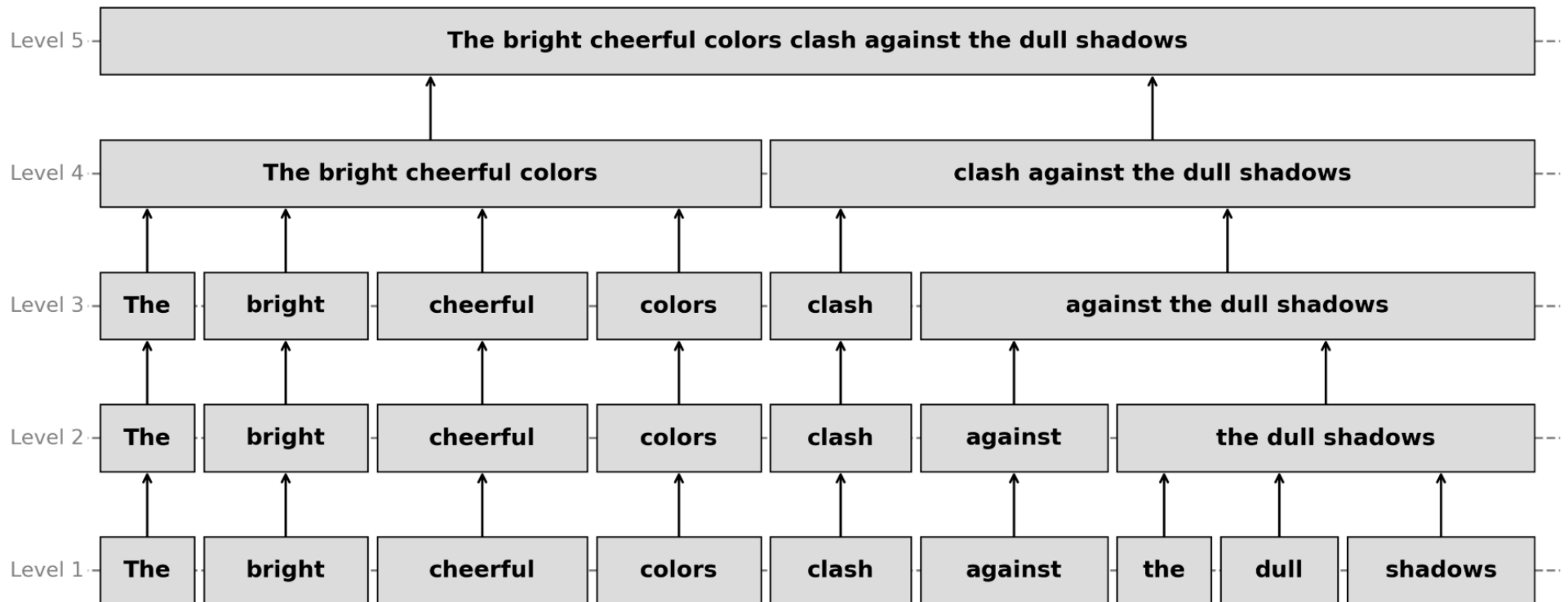


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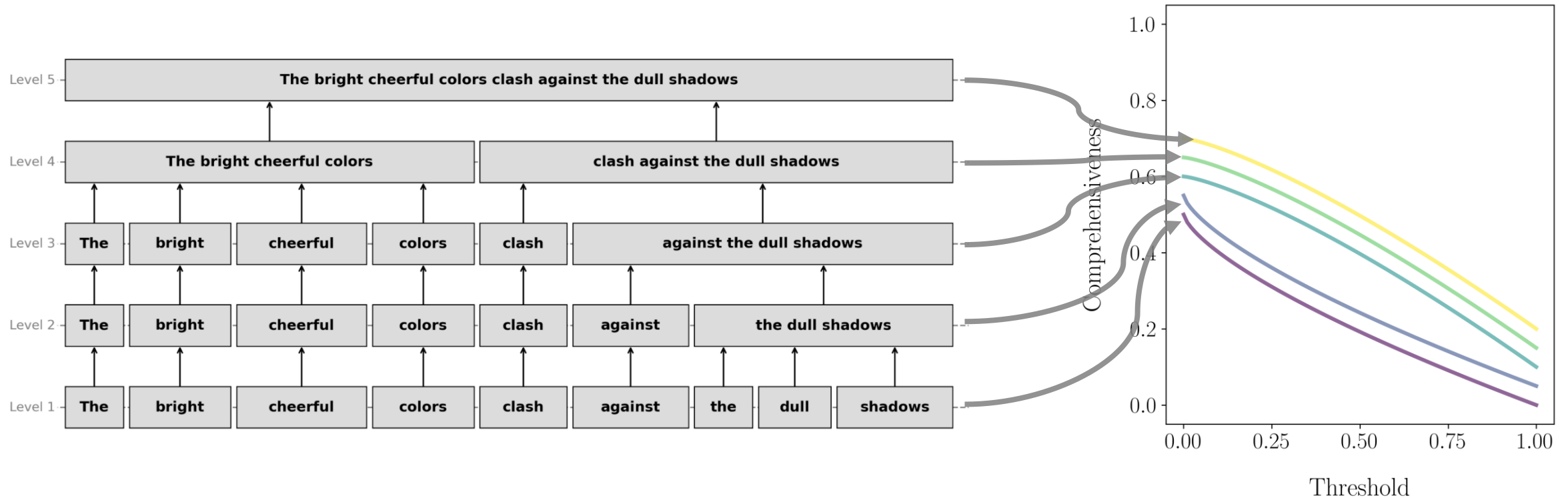


EVALUATION METRICS

High-order Attributions

We will evaluate high-order attributions from the **faithfulness** perspective.

Step 2: construct {comprehensiveness, sufficiency}-threshold curves for each level.



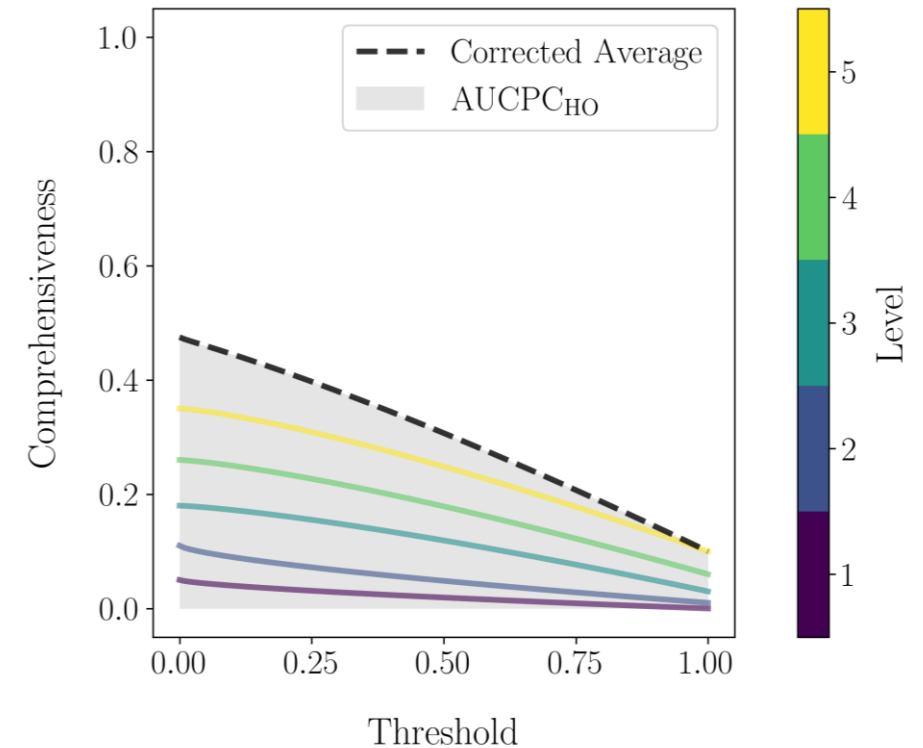
EVALUATION METRICS

High-order Attributions

We will evaluate high-order attributions from the **faithfulness** perspective.

Step 3: compute a global {comprehensiveness, sufficiency}-threshold curve, correcting for:

- **Sentence length bias:** Levels closer to the root may be underrepresented due to shorter sequences that do not reach such heights.
- **Span coverage bias:** nodes at higher levels in constituency trees correspond to longer textspans, which are more likely to receive inflated attribution scores



EVALUATION METRICS

Global Qualitative Evaluation

Following previous studies on global explainability for text, we will use **wordclouds** to visualize the globally most important words for the model.

- ↳ To assess **higher-order** importance, we will create **wordclouds for 2-grams** (pairs of words).

IDG APPLIED TO HATE SPEECH DETECTION

Dataset

Model Selection

Explainability

Disclaimer about hateful and offensive content

This work involves the use of the HateXplain dataset, which contains real-world examples of hate speech and offensive language. Such instances cannot be avoided due to the nature of the work. As a result, some examples included in this presentation contain explicit, offensive, or harmful content. These excerpts are presented for the unique purpose of scientific analysis and do not reflect the views of the author or affiliated institutions. Viewer discretion is advised.

IDG APPLIED TO HATE SPEECH DETECTION

- **Dataset:** HateXplain dataset.
 - ↳ **Origin:** text sequences extracted from Twitter and Gab.
 - ↳ **Labels:** normal / hate speech.
 - ↳ **Annotations:** hateful sequences are annotated with human rationales indicating the words that support the classification.

IDG APPLIED TO HATE SPEECH DETECTION

- **Dataset:** HateXplain dataset.

Split	Class		Total	Ratio (N:HS)
	Normal	Hate Speech		
Train	6251	4748	10 999	1.32
Validation	781	593	1374	1.32
Test	782	594	1376	1.32
Total	7814	5935	13 749	1.32

IDG APPLIED TO HATE SPEECH DETECTION

- **Model selection:** BERT vs. XLNet with identical fine-tuning strategies.
 - ↳ **Base model:** bert-base-uncased / xlnet-base-cased (HuggingFace).
 - ↳ **Batch size:** 32.
 - ↳ **Learning Rate:** 2×10^{-5} , with linear scheduling.
 - ↳ **Optimizer:** AdamW, weight decay 0.01.
 - ↳ **Early Stopping:** max. 5 epochs, monitoring validation loss.

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PERFORMANCE ON THE VALIDATION SPLIT

Model	Metric				
	Accuracy	Precision	Recall	F ₁ -score	AUROC
BERT	0.878	0.855	0.865	0.860	0.948
XLNet	0.870	0.862	0.833	0.847	0.937

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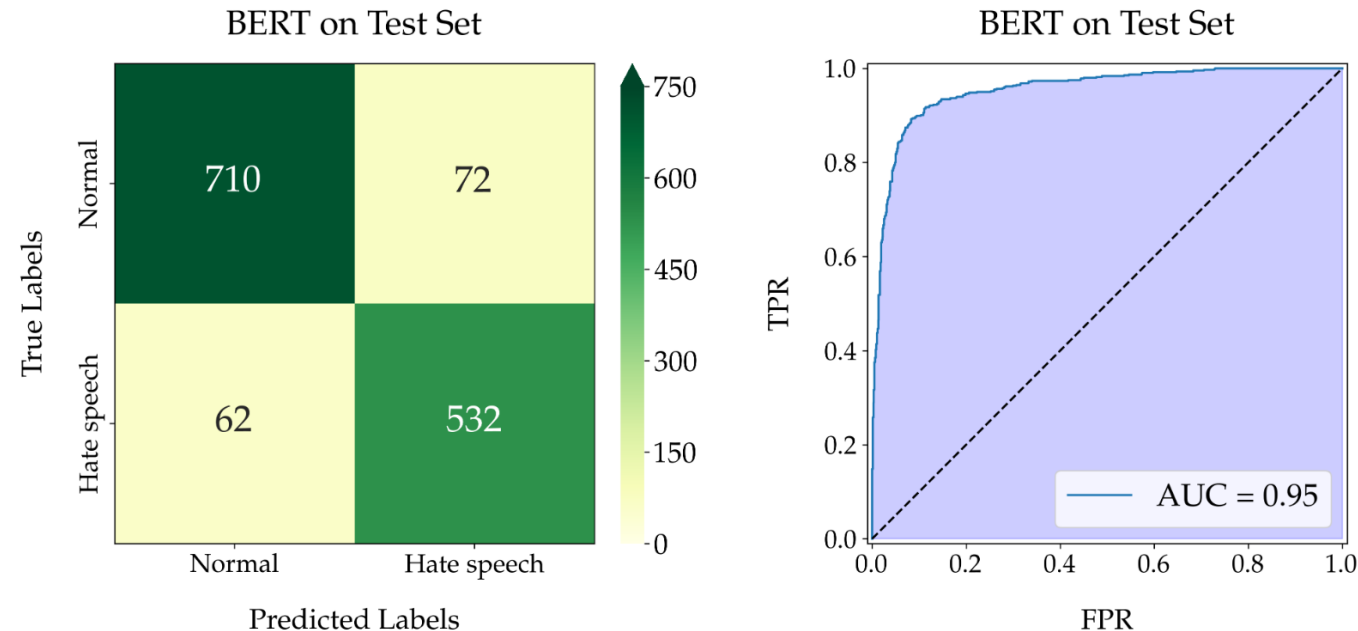
BERT was selected due to its higher ability to detect the positive (hate speech) class

IDG APPLIED TO HATE SPEECH DETECTION

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PERFORMANCE ON THE TEST SPLIT

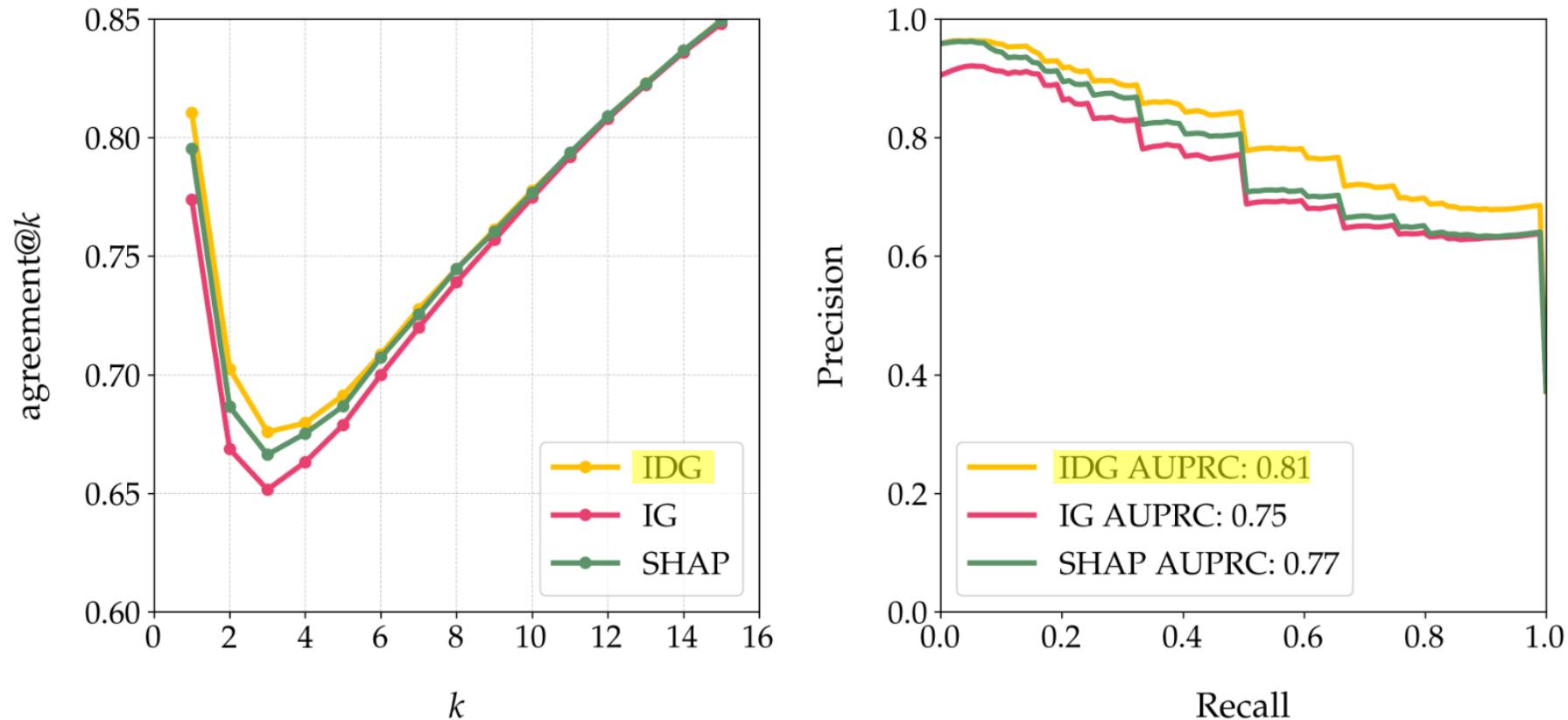
Model	Metric				
	Accuracy	Precision	Recall	F ₁ -score	AUROC
BERT	0.903	0.881	0.896	0.888	0.954



IDG APPLIED TO HATE SPEECH DETECTION

- **Explainability:** IDG vs. IG vs. SHAP on word-level attributions.

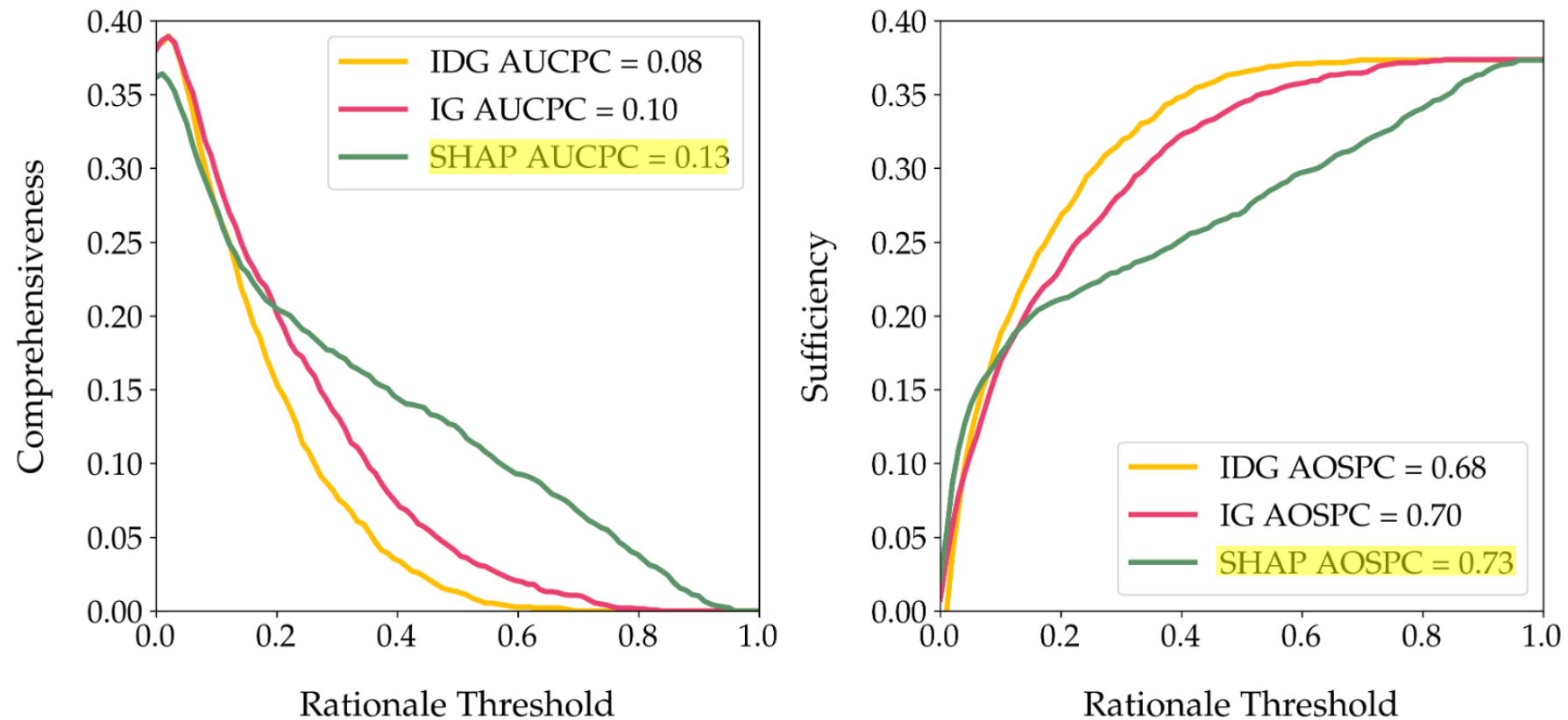
WORD-LEVEL PLAUSIBILITY



IDG APPLIED TO HATE SPEECH DETECTION

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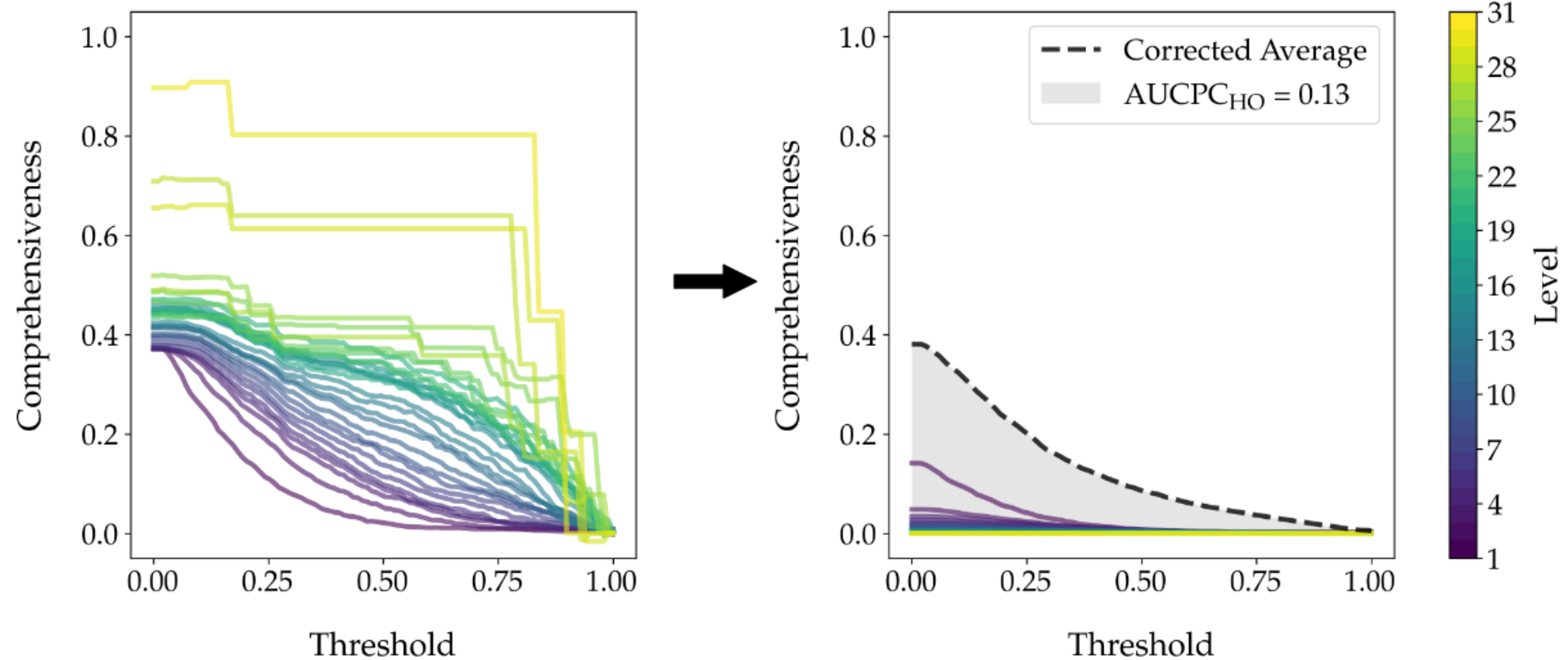
WORD-LEVEL FAITHFULNESS



IDG APPLIED TO HATE SPEECH DETECTION

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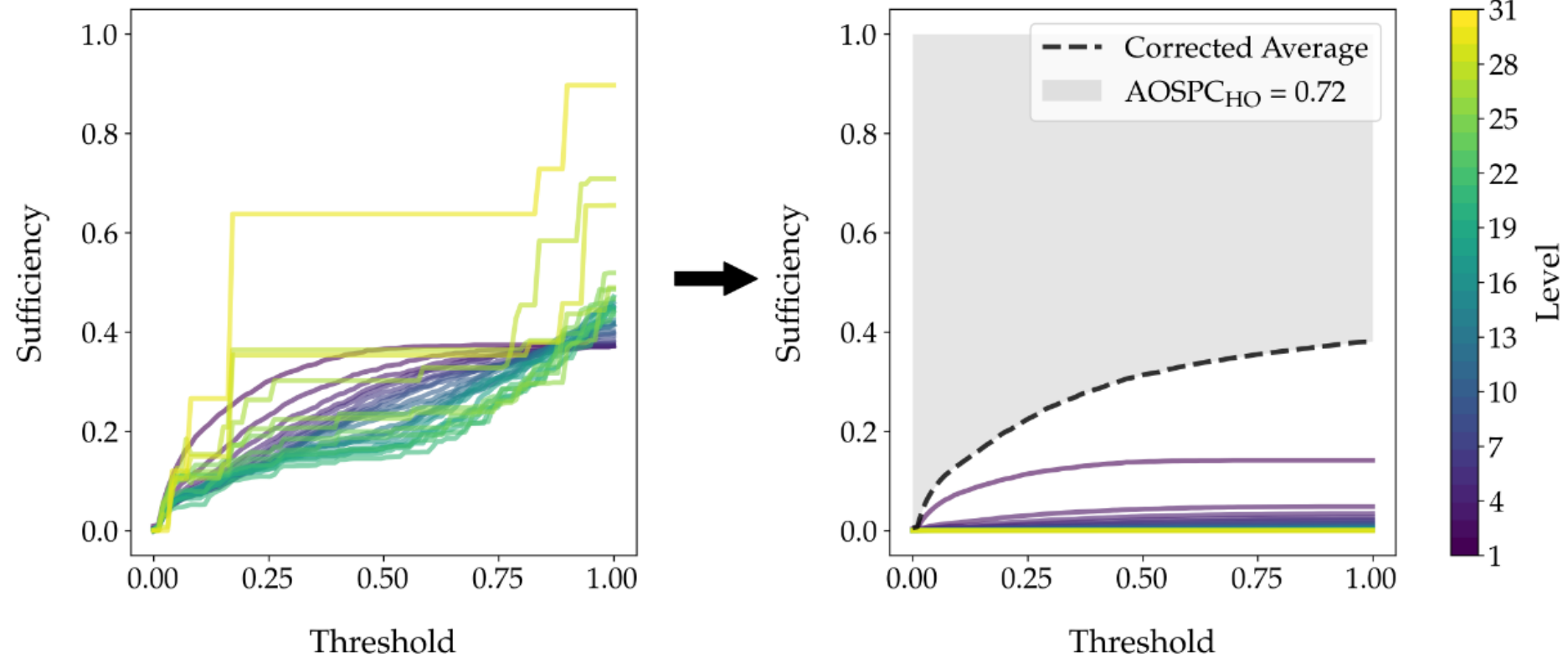
HIGH-ORDER COMPREHENSIVENESS



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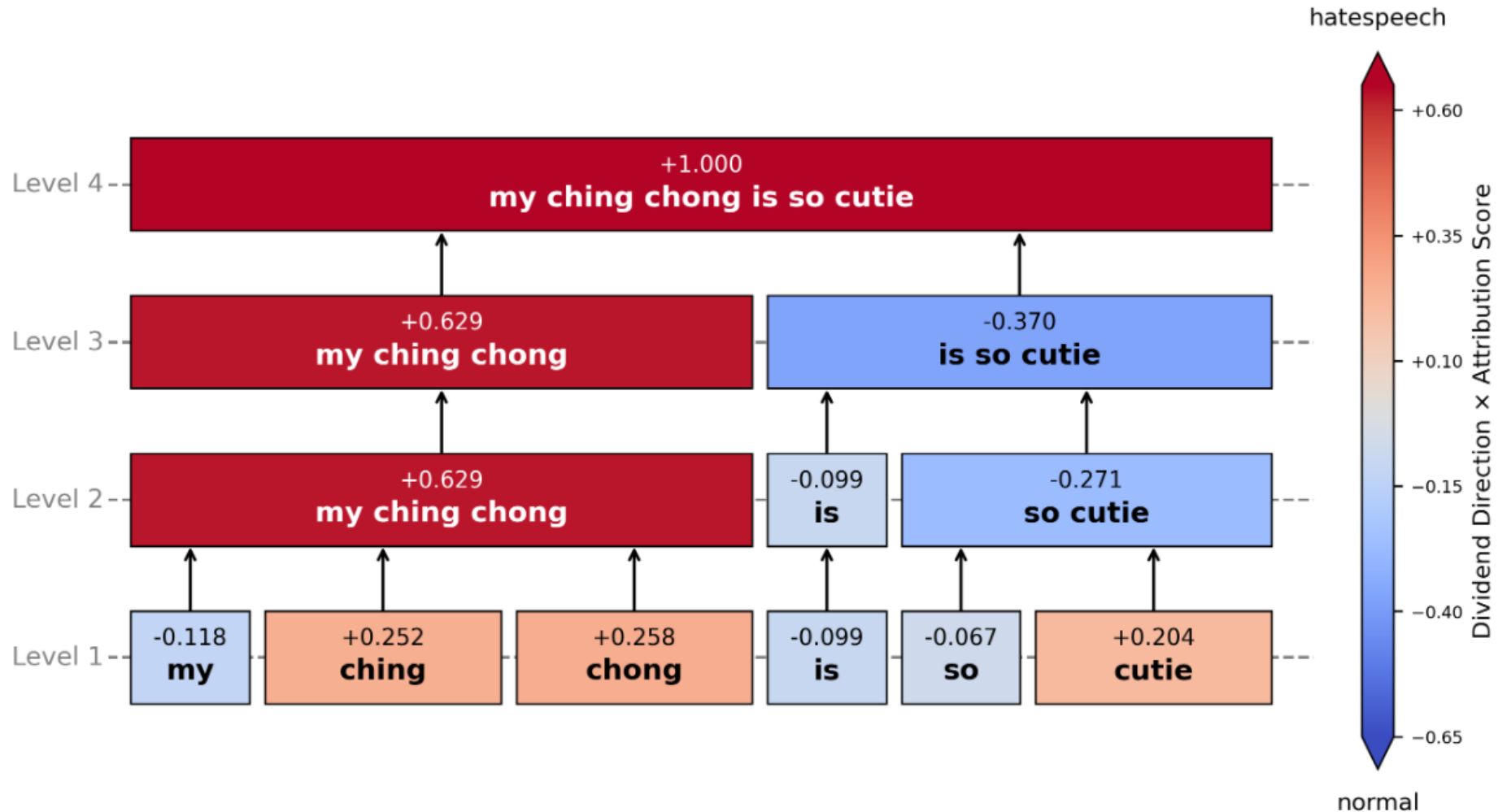
- **Explainability:** IDG on high-order attributions.

HIGH-ORDER SUFFICIENCY



IDG APPLIED TO HATE SPEECH DETECTION

- **Explainability:** example of IDG output.



IDG APPLIED TO HATE SPEECH DETECTION

- **Explainability:** global qualitative insights.

IDG

IG

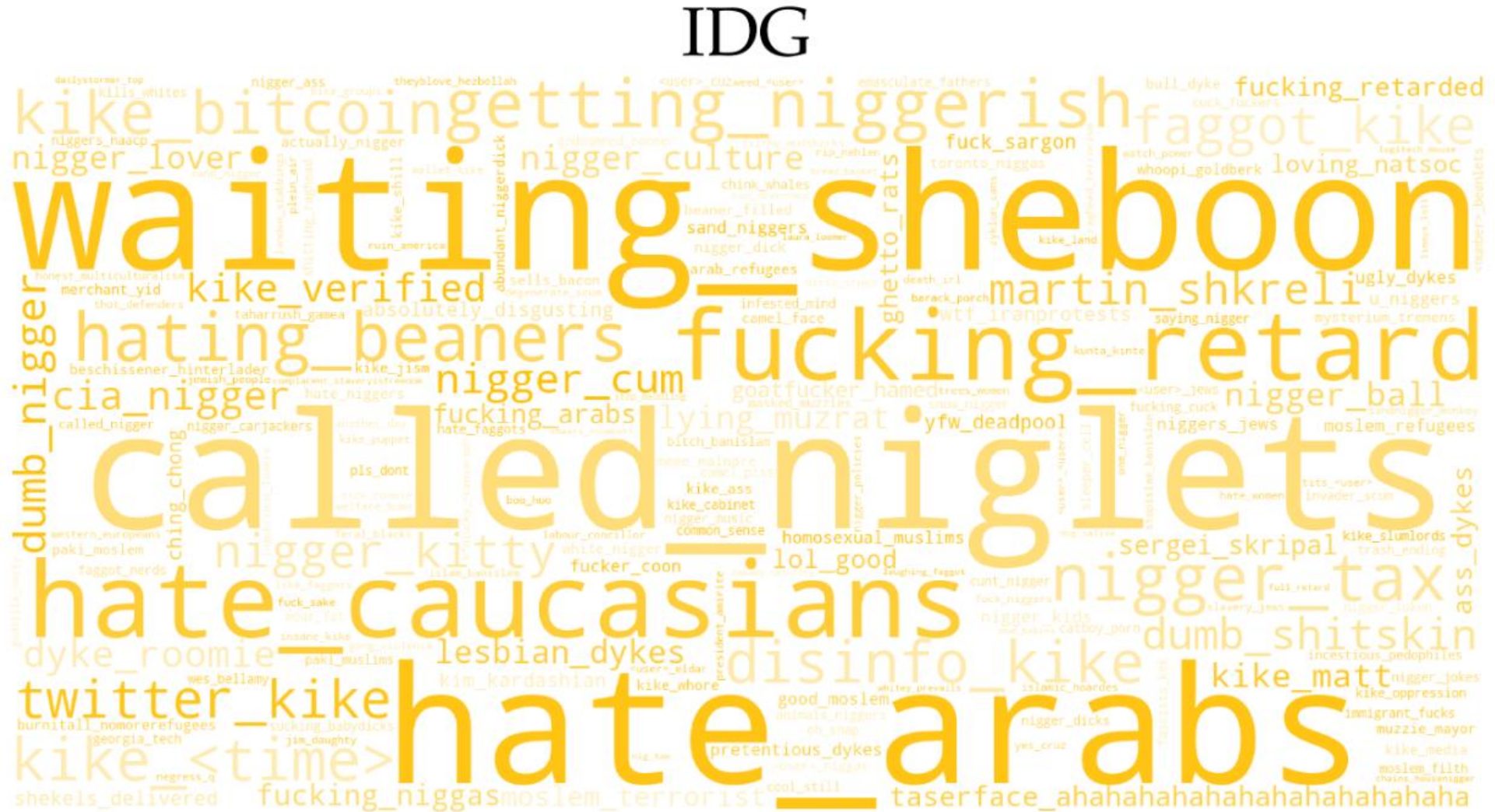


SHAP



IDG APPLIED TO HATE SPEECH DETECTION

- **Explainability:** global qualitative insights.



CONCLUSIONS

CONCLUSIONS

Conclusions: key insights on IDG for text model explainability

- Shows higher plausibility than IG and SHAP, aligning better with human intuition in detecting hateful content.
- Shows lower word-level faithfulness, but this improves with high-order interaction modeling.
- Demonstrates the importance of high-order attributions for capturing complex model reasoning.
- IDG provides interpretable and meaningful explanations aligned with toxic discourse patterns.

Overall, IDG offers a more comprehensive understanding of the model's decision-making process.

Explaining Word Interactions Using Integrated Directional Gradients

Author: Marc Ballesteró Ribó

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- Dr. Daniel Ortiz Martínez



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Thanks for your attention!

FUTURE WORK

Future Work

- Plausibility metrics depend on subjective word-level human annotations. Data from a larger pool of annotators would validate the analysis.
- Faithfulness metrics use perturbations that may create unrealistic, out-of-distribution inputs. Causal, counterfactual or generative approaches could be explored.
- Evaluation limited to a single dataset. Broader testing needed for generalizability.
- No user studies conducted, need to assess interpretability across different audiences.
- Multilingual analysis. Future work should explore other languages and cultures.
- Text-only data. multimodal content (links, audio, video) remains unexplored.

INTEGRATED DIRECTIONAL GRADIENTS

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(1) Non-negativity: the importance of any feature subset should be non-negative.

$$\forall S \in M, v(S) \geq 0$$

↳ Feature importance should be a proxy of the amount of information (statistical entropy) that is used by the model $\Rightarrow$ Non-negative quantity.

INTEGRATED DIRECTIONAL GRADIENTS

Axiomatic Requirements for a Solution of the Feature Group Attribution Problem

(2) Normality: the importance of the empty set of features should be zero.

$$v(\emptyset) = 0$$

↳ No information means no importance.

INTEGRATED DIRECTIONAL GRADIENTS

Axiomatic Requirements for a Solution of the Feature Group Attribution Problem

(3) Monotonicity: the importance of a set of features is greater than or equal to the importance of its subsets.

$$\forall S, T \in M, S \subseteq T \implies v(S) \leq v(T)$$

↳ Adding more features means adding more information, thus more feature importance should be attributed.

INTEGRATED DIRECTIONAL GRADIENTS

Axiomatic Requirements for a Solution of the Feature Group Attribution Problem

(4) Superadditivity: the value of the disjoint union of two feature subsets is greater than or equal to the sum of values of the two subsets.

$$\forall S, T \in M, S \cap T = \emptyset, v(S \sqcup T) \geq v(S) + v(T)$$

↳ By hypothesis, feature collaboration cannot be detrimental. Combined importance attribution should be larger.

INTEGRATED DIRECTIONAL GRADIENTS

Axiomatic Requirements for a Solution of the Feature Group Attribution Problem

(5) Sensitivity (a): if a feature x_i is such that $f(x) \neq f(b)$ for every input and baseline that only differ in x_i , then

$$v(\{x_i\}) > 0 \text{ and } v(S) > 0$$

for every set of features containing the value of the i -th feature.

↳ A feature that does effectively affect the output of the predictor should be attributed a nonzero importance

INTEGRATED DIRECTIONAL GRADIENTS

Axiomatic Requirements for a Solution of the Feature Group Attribution Problem

(6) Sensitivity (b): if a feature x_j is such that $f(x) = f(b)$ for every input and baseline that only differ in x_j , then

$$v(\{x_j\}) = 0 \text{ and } v(S) = v(S \setminus \{x_j\})$$

for every set of features containing the value of the j -th feature.

↳ Any feature irrelevant for the predictor is attributed a zero importance, and it does not alter the importance of any feature group containing it

INTEGRATED DIRECTIONAL GRADIENTS

Axiomatic Requirements for a Solution of the Feature Group Attribution Problem

(7) Symmetry preservation: if two features x_i, x_j are both *functionally and structurally equivalent*, then given an input such that $x_i = x_j$ and a baseline such that $b_i = b_j$, then

$$v(S \cup \{x_i\}) = v(S \cup \{x_j\})$$

for every set of features not containing the values of the i, j -th features.

- ↳ Features that are both
 - functionally equivalent $\rightarrow$ swapping the values of those features does not alter the output
 - structurally equivalent $\rightarrow$ occupy equivalent positions in the family of meaningful subsets
- apport the same amount of importance to all feature subsets there may be included in.

INTEGRATED DIRECTIONAL GRADIENTS

Axiomatic Requirements for a Solution of the Feature Group Attribution Problem

- (8) Implementation invariance:** given two DNN predictors f_1, f_2 such that $f_1(x) = f_2(x), \forall x \in \mathcal{X}$, the method attributes equivalent importance scores to both of them.
- ↳ The attribution of importance scores does not depend on the DNN architecture, as long as the predictions are the same.

IDG APPLIED TO HATE SPEECH DETECTION

- **Explainability:** example of IDG output.

